

Anurag Mishra

Medium: anuragmishra.27746

Github: github.com/anurag-mishra899

Email: anurag.mishra899@gmail.com

Mobile: +91-9456243057

LinkedIn: in/anurag-mishra-660961b7

PROFILE SUMMARY

- **Experience:** 6.5+ YoE in designing and developing **Machine Learning, Deep Learning, Generative AI, and NLP** solutions for various industries, including retail, banking, telecommunications, and chemicals
- **Frameworks:** PyTorch, Langchain, LlamaIndex, Fastapi, Scikit, NLTK, SpaCy, TensorFlow, Keras
- **Tools:** Kubernetes, Docker, GIT, PostgreSQL, MySQL, SQLite
- **Platforms:** Azure Cloud, AWS, GCP, Linux
- **Certified in:** Azure Fundamental, Azure Data Scientist, Databricks ML Professional
- **AI Conference as Speaker:** LLM Application, Agentic Workflow, Multi Modal LLMs
- **Blogs/Publications:** Article 1, Article 2, Article 3, Article 4

EXPERIENCE

- **EY GDS** Hybrid
Associate Manager (Full-time) Feb 2022 - Present
 - **Deployed GenAI Chat-bot Solution for 40 Branches, Serving 2,000-3,000 Users:**
 - * **Led a team of six engineers** to design and deploy a Chat-bot solution for **2,000-3,000 users** across **40 branches** of the largest bank in the Philippines.
 - * **Reduced query resolution time** from 5-10 minutes to under 30 seconds, **boosting customer satisfaction by 45%.**
 - **Developed LLM Production Pipeline, Reducing Downtime by 50%:**
 - * Engineered a custom preprocessing and chunking pipeline to **manage complex tables, PDF formatting, and flowcharts**, increasing retrieval performance by **55%.**
 - * Built a scalable deployment pipeline using FastAPI, Langchain, Azure API Gateway, and Kubernetes, **supporting throttling, fallbacks, and region-wise load balancing OpenAI models**, reducing downtime by **50%.**
 - **Implemented LLM Ops and Evaluation Pipeline, Boosting Accuracy by 46%:**
 - * Introduced **Custom RAG evaluation framework** for automated end-to-end assessment, enhancing performance and optimizing solution components such as routing, retrieval, and generation, resulting in a **46% increase in accuracy.**
 - * Developed a monitoring pipeline using **Langfuse for observability and logging**, reducing issue resolution time by **30%.**
 - * Established **feedback mechanism**, achieving a **25% improvement** in solution accuracy over six months.
 - **Smart Call-Centre Solution for US-Based Client, Increasing Agent Productivity by 60%:**
 - * Built LLM and data ingestion pipeline integrating Azure DevOps, Langchain, and ServiceNow, managing post-call actions including summarization, ticket creation, updates, and quality assurance.
 - * Designed **pipeline to handle 300-400 calls per center**, increasing agent productivity by **60%** and reducing average call handling time by **40%.**
 - **Fine-tuning Transformer Models Reducing Operational Costs by 70%:**
 - * Delivered advanced techniques in deep learning model tuning, leading to significant improvements in **automation extraction workflows** across the organization; measured performance gains resulted in reduced processing times of critical data inputs by over **80% hours** monthly.

- * Deployed deep learning models as **batch endpoints utilizing Azure Web Apps, Databricks Jobs, and MLflow**, ensuring streamlined integration and efficient model management, reducing operational costs by 70%.

- o **Executed Go-to-Market Strategy, Increasing Client Engagement by 35%:**

- * Contributed to the "Go to Market" strategy for Generative AI across various clients, developing prototypes for industry use-cases using models like OpenAI, LLaMa, MPT, and Dolly, resulting in a 35% increase in client engagement.
- * Conducted client awareness sessions on **"Industry Implementation and Risks of GenAI," educating over 200 clients** on the practical applications and potential risks of generative AI technologies.
- * Successfully drove the adoption of Generative AI solutions in **three major client projects, increasing revenue by \$500K.**

- **Deloitte (Office of US)**
Consultant - AI (Full-time)

Remote
Feb 2021 - Feb 2022

- o **Fixture Optimization for Inventory Management, Achieved \$5 Million Cost Reduction:** Devised optimization solution to allocate the optimal number of fixtures for each product and brand in retail chain stores, resulting in a \$5 million reduction in costs and expenses.
- o **Sales Forecasting for Retail Client, Enhanced Demand Forecasting by 45%:** Developed sales forecasting model for each department per store to assess demand. Utilized tree-based algorithms such as Gradient Boosting, XGBoost, and LightGBM, as well as regression-based models like ARIMA & SARIMA.

- **Virtusa Polaris**
Machine Learning Engineer (Full-time)

Hybrid
June 2018 - Feb 2021

- o **Multi-Class classification of Trading Products, Reducing Manual Tagging from 15k/hour to 1 Million/hour:** Devised solution for Product Controllers to reduce person-hours spent on tasks of identifying breach in trading activity.
- o **NLP Models to Assess Commentary using SpaCy, Improving Comments Quality by 80% :** Developed NLP solution to identify the inconsistency in PnL commentary generated by Product Controllers. Milestones included building Rest API to render user request and ETL operations to handle user requests.
- o **ML Models & Techniques:** Utilized various ML algorithms like Random Forest, Logistic regression Gradient Boosting and Hyper-parameter tuning to get 95% accuracy.
- o **Tools & Technologies:** Used Python programming and worked on libraries like Pandas, Scikit-learn, NumPy, Matplotlib, Seaborn, Spacy

PROJECTS

- **Fine-Tuned LLM for Domain Specific tasks using LoRA, Increasing Domain-based NER By 55%:** Performed Fine-tuning of LLM models(LLaMA 3.1 8B, Mistral 7B) on 80 GB A-100 GPU on custom build dataset for Manufacturing domain.
- **LLM Based Agentic flow for Text to Code Solution:** Agent based solution that convert requirements to code. The solution uses multi-agents (debugger, coder, tester etc) to write the bug-free and optimized code.

EDUCATION

- **National Institute of Technology, Manipur**
Bachelor of Technology - Information Technology; GPA: 8.5
Courses: Operating Systems, Data Structures, Analysis Of Algorithms, Artificial Intelligence, Machine Learning, Networking, Databases

India

July 2014 - June 2018

HONORS AND AWARDS

- Awarded Excellence Client Performance Award in EY
- Honored to be among top 10 students in CSE Department
- Achieved (Among Top 1 % in board) in the entire district in HSC exam.